

A comparative study on viscosity of human, bovine and pig IgG immunoglobulins in aqueous solutions

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Abstract

This paper presents the results of viscosity determinations on aqueous solutions of human, bovine and pig IgG immunoglobulins over a wide range of concentrations and at temperatures ranging from 5°C to 55°C. On the basis of the generalized Arrhenius formula, the viscosity–temperature and the viscosity–concentration dependence of the solutions are discussed. By applying an asymptotic form of the generalized Arrhenius formula, such rheological quantities as the intrinsic viscosity and Huggins coefficient were calculated. © 1999 Elsevier Science B.V. All rights reserved.

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1. Introduction

Immunoglobulins comprise a heterogeneous family of proteins with antibody activity and constitute about 20% of total plasma proteins. Of the five classes of immunoglobulins, designated IgG, IgA, IgM, IgD and IgE, the IgG immunoglobulins constitute approximately 75% of the total serum immunoglobulins [1]. IgG, which are investigated in the present paper, can be further divided into subclasses on the basis of relatively minor antigen-binding specificities. However, the gross conformation of the IgG molecules in all subclasses is the same. The basic unit of all IgG is a four polypeptide chain structure comprising two identical light chains and two identical heavy chains which are held together by non-covalent forces and usually by covalent inter-chain disulfide bridges so that the structure looks like a letter T or Y [1–3]. The four polypeptide chains are folded in three well-defined globular structures, two F_{ab} fragments and one F_c fragment, connected by a single joint which acts as a semiflexible hinge, allowing internal motions of the fragments. The inherent flexibility of IgG has been investigated using the fluorescence anisotropy decay method [4–6]. Some sophisticated theoretical methods have also been applied to the problem

in the literature [7,8]. However, hydrodynamic properties such as sedimentation, diffusion coefficients or viscosity depend essentially on the overall conformation of the macromolecules and are not very sensitive to segmental flexibility.

The hydrodynamic properties of immunoglobulins in solution have not yet been sufficiently studied. This is especially the case for the viscosity of immunoglobulin solutions, for which only the data for very low concentrations are available and only at 25°C [9,10]. In the present study, the results of viscosity determinations on aqueous solutions of human, bovine and pig IgG over a wide range of concentrations and at temperatures ranging from 5 to 55°C are presented. A comparison of parameters from a generalized Arrhenius formula [11] connecting the viscosity with temperature and concentration of the dissolved proteins is presented. At low concentrations, the intrinsic viscosity and the Huggins coefficient from the generalized Arrhenius formula are determined.

2. Materials and methods

Highly purified bovine and pig IgG were purchased from Sigma and human IgG from Polish Chemical Reagents factories and were used without further purifi-

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cation. From the crystalline form the material was dissolved in distilled water and the solution was treated to remove dust particles with filter papers. The samples were stored at 4°C until just prior to viscometry measurements, when they were warmed from 5 to 55°C. The pH values of prepared samples were close to their isoelectric points and were as follows: 5.6 for human IgG, 5.65 for bovine IgG and 5.9 for pig IgG. These values varied slightly with different protein concentrations.

Viscosity was measured with a Ubbelohde-type capillary microviscometer with a flow time for water of 28.5 s at 25°C. The microviscometer was placed in a water-bath controlled thermostatically at 5°C to 55 ± 0.05°C. The same viscometer was used for all measurements and was mounted so that it always occupied the same position in the bath. Flow times were recorded to within 0.1 s. Solutions were temperature-equilibrated and passed once through the viscometer before any measurements were made. For most concentrations the viscosity measurements were made from 5°C to 55°C in 5°C intervals. This range of temperatures was chosen because above 55°C the thermal denaturation of IgG occurs [12] and the proteins show a highly pronounced tendency to aggregate [13,14]. This heat-induced aggregation reverses upon cooling.

Solutions densities were measured by weighing. Protein concentrations were determined by a dry weight method in which samples were dried at high temperature for several hours. The viscosities were measured for concentrations from 16.92 kg/m³ up to 284.9 kg/m³ for human IgG, from 8.8 kg/m³ to 207.7 kg/m³ for bovine IgG and from 28.4 kg/m³ to 223.3 kg/m³ for pig IgG.

3. Results and discussion

Very recently we have demonstrated that the most useful relation connecting the viscosity η with temperature is a somewhat modified Arrhenius formula [11,15]. It describes the viscosity–temperature dependence from the neighbourhood of solution freezing-point up to the vicinity of the temperature at which thermal denaturation of the protein occurs. The relation has the form:

$$\eta = \exp\left(-B + DT + \frac{E_s}{RT}\right) \quad (1)$$

where B and D are parameters, E_s is the activation energy of viscous flow of solution, R is the gas constant and T is absolute temperature. Fig. 1 shows the temperature dependence of the viscosity of human, bovine and pig IgG aqueous solutions for the high concentrations. As seen, curves obtained by using the relation from Eq. (1) give a very good fit to the experimental points over the whole range of temperatures. For the smaller con-

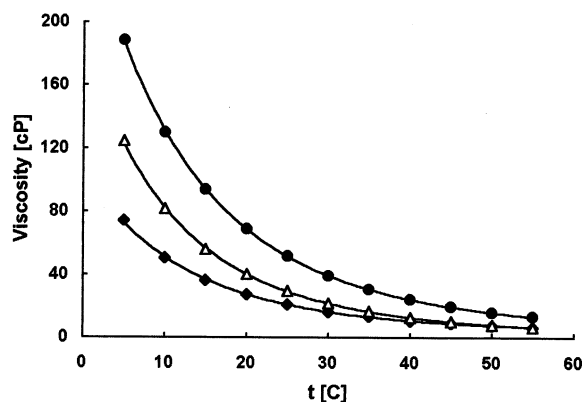


Fig. 1. Temperature dependence of the viscosity of human IgG (●), bovine IgG (△) and pig IgG (◆) aqueous solutions for concentrations $c = 263.2, 200.3$ and 223.3 kg/m³, respectively. The curves show the fit obtained by using Eq. (1) with the parameters: $B = 57.011$, $D = 6.257 \times 10^{-2} \text{ K}^{-1}$ and $E_s = 87.727 \text{ kJ/mol}$ for human IgG; $B = 67.788$, $D = 7.674 \times 10^{-2} \text{ K}^{-1}$ and $E_s = 102.56 \text{ kJ/mol}$ for bovine IgG; $B = 78.183$, $D = 9.881 \times 10^{-2} \text{ K}^{-1}$ and $E_s = 111.19 \text{ kJ/mol}$ for pig IgG.

centrations the situation is the same. For all concentrations the parameters B , D and E_s were calculated by using the least squares method as we have previously described [11]. The experimental values of water viscosity (when given in centipoise) agree very well with those calculated on the basis of Eq. (1) when $B = B_w = 25.936$, $D = D_w = 2.014 \times 10^{-2} \text{ K}^{-1}$ and $E_s = E_w = 32.013 \text{ kJ/mol}$.

Viscosity measurements for bovine serum albumin [11] and lysozyme [15] aqueous solutions showed that the activation energy of viscous flow E_s is a monotonically increasing function of concentration. Fig. 2 shows that this is the case for all investigated IgG too. As we have shown in the above-mentioned paper [11], the energy activation–concentration dependence may be written in the following form:

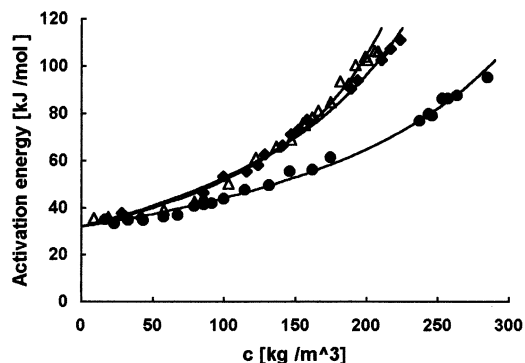


Fig. 2. Plot of the solution activation energy E_s versus concentration c , showing mean experimental points for human (●), bovine (△) and pig (◆) IgG, respectively. The curves show the fit according to Eq. (2) with parameters from Table 1.

Table 1

Molecular masses and the parameters of the investigated IgG samples obtained from the fit of Eqs. (2)–(4) to the experimental points

	M_p (kDa)	v (m ³ /kg)	B_p	D_p (K ⁻¹)	E_p (kJ/mol)
Human IgG	156	2.060×10^{-3}	4.667×10^5	688.53	8.285×10^5
Bovine IgG	160	3.283×10^{-3}	6.545×10^5	861.22	1.192×10^6
Pig IgG	160	2.599×10^{-3}	9.031×10^5	1404.37	1.361×10^6

$$E_s = \frac{c}{\alpha - \beta c} (E_p - E_w) + E_w \quad (2)$$

where $\alpha = \rho_w M_p / M_w$ and $\beta = \alpha v - 1$. The quantities c , ρ_w , v , M_p and M_w denote the solute concentration and water density in kg/m³, the effective specific volume of a protein and the molecular masses of the dissolved proteins and water, respectively. The physical meaning of E_s and v was presented in our earlier works [11,15]. The molecular mass of human IgG is 156 kDa and in this case $\alpha = 8.667 \times 10^6$ kg/m³. For bovine and pig IgG $M_p = 160$ kDa and $\alpha = 8.889 \times 10^6$ kg/m³. In Eq. (2) the activation energy of dissolved proteins E_p and the effective specific volume of a particle v must be taken into account as two adjustable parameters. The parameters were also calculated by using the least squares method. The obtained values are summarized in Table 1. As seen in Fig. 2, Eq. (2) then gives good approximation to the experimental values.

The coefficients B and D from Eq. (1) depend on the concentration in exactly the same way as E_s . Both B and D increase monotonically with increasing concentration. Therefore, one can write the same relations as for E_s :

$$B = \frac{c}{\alpha - \beta c} (B_p - B_w) + B_w \quad (3)$$

$$D = \frac{c}{\alpha - \beta c} (D_p - D_w) + D_w \quad (4)$$

where v , B_p and D_p are adjustable parameters. The above relations give a good fit to the experimental values for parameters listed in Table 1.

One can easily obtain the concentration dependence of relative viscosity $\eta_r = \eta / \eta_0$, where η and η_0 are the viscosities of the solution and the solvent, respectively. By substituting Eqs. (2)–(4) into Eq. (1), one can obtain the following formula for relative viscosity of solution:

$$\eta_r = \exp \left\{ \frac{c}{\alpha - \beta c} \left[-(B_p - B_w) + (D_p - D_w)T + \frac{E_p - E_w}{RT} \right] \right\} \quad (5)$$

At a fixed temperature the above formula describes the viscosity–concentration dependence of a solution. Fig. 3 shows the experimental points for all investigated IgG at the temperature 25°C and the curves resulting from Eq. (5). As has been shown in our earlier work

[16], the relative viscosity of aqueous solutions of globular proteins depends strongly on the axial ratio of the dissolved proteins. We proved that the higher the axial ratio of dissolved proteins the higher the relative viscosity for a given concentration. Human IgG molecules in solution behave equivalently as prolate ellipsoids of revolution with the axial ratio $p = 5.34$. As seen in Fig. 3, the relative viscosity for human and pig IgG increases with increasing concentration in nearly exactly the same way. Thus, one can conclude that human and pig IgG molecules in solution behave as prolate ellipsoids of revolution with the same axial ratio, which suggests that they have nearly exactly the same shape. Fig. 3 shows that bovine IgG molecules, in turn, behave as prolate ellipsoids of revolution with a somewhat higher axial ratio or that they are more aspheric. In fact, it suggests that the average angle between F_{ab} fragments for bovine IgG is somewhat smaller than for human and pig ones.

The values of activation energy, which we previously obtained for water ($E_w = 32.013$ kJ/mol), lysozyme ($E_p = 3.97 \times 10^4$ kJ/mol), bovine serum albumin ($E_p = 5.374 \times 10^5$ kJ/mol) and now for IgG (Table 1), show that this quantity depends strongly on the molecular weights and dimensions of the molecules. Differences in the activation energy for human and pig IgG, which have nearly the same molecular weight and dimensions, suggest that some parameters related to the protein–protein or/and protein–solvent interactions may be important too. As is seen in Table 1 the parameters v , B_p and D_p are even more differentiated than the activation energy of the investigated proteins. An explanation of the problem needs more experimental data and some theoretical progress in the field is also necessary.

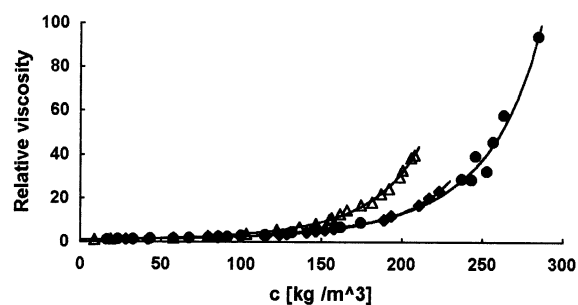


Fig. 3. Plot of the relative viscosity η_r versus concentration c for human IgG (●), bovine IgG (△) and pig IgG (◆). The curves show the fit obtained by using Eq. (5) with parameters from Table 1.

Table 2

The numerical values of the intrinsic viscosity $[\eta] \times 10^3$ (m³/kg) for all investigated IgG calculated from Eq. (7)

	<i>t</i> (°C)										
	5	10	15	20	25	30	35	40	45	50	55
Human	9.591	9.258	8.951	8.667	8.407	8.168	7.949	7.751	7.572	7.409	7.265
Bovine	11.29	10.76	10.25	9.782	9.344	8.936	8.558	8.206	7.882	7.582	7.306
Pig	8.538	8.159	7.821	7.521	7.258	7.029	6.834	6.669	6.536	6.430	6.352

Eq. (5) describes the viscosity–concentration dependence (for a given temperature) from very diluted up to extremely concentrated solutions and allows us to calculate, in a very convenient way, the intrinsic viscosity and the Huggins coefficient. As is well known, at low concentrations, the relation between the solution viscosity and the concentration may be expressed by the polynomial [17]:

$$\frac{\eta_{sp}}{c} = [\eta] + k_1[\eta]^2c + k_2[\eta]^3c^2 + \dots \quad (6)$$

where $[\eta] = \lim_{c \rightarrow 0} \frac{\eta_{sp}}{c}$ is the intrinsic viscosity, $\eta_{sp} = \eta_r - 1$ is the specific viscosity and the dimensionless parameters k_1 , k_2 are coefficients of expansion; k_1 is called the Huggins coefficient. In our case, the generalized Arrhenius formula (Eq. (5)) can be expanded in the power series of concentration. An expansion identical to that in Eq. (6) can be obtained from:

$$[\eta] = \frac{1}{\alpha} \left[-(B_p - B_w) + (D_p - D_w)T + \frac{E_p - E_w}{RT} \right] \quad (7)$$

$$k_1 = \frac{1}{2} \left[\frac{2\beta}{-(B_p - B_w) + (D_p - D_w)T + \frac{E_p - E_w}{RT}} + 1 \right] \quad (8)$$

The higher coefficients of expansion have a more complicated form and are omitted here. The traditional way for treating viscosity data at low concentrations consists of plotting η_{sp}/c against concentration, extrapolating it to the intercept (equal to $[\eta]$) and obtaining the coefficient k_1 from the corresponding slope. However, even if $\eta_{sp} < 0.7$ the concentration dependence of η_{sp}/c is curved, so that linear extrapolation gives a serious error in $[\eta]$ and k_1 [17]. The new method pre-

sented above was successfully applied for bovine serum albumin [11], lysozyme [15], mammalian haemoglobin [18] and now for IgG. The accordance of some results obtained by the new method with those accessible in the literature shows that it is at least as precise as the traditional method and is simple and convenient. Moreover, it seems that this method is the only reasonable one which allows calculation of the higher coefficients of expansion k_2 , k_3 and so on. The traditional method gives enormous errors in this case.

As seen, both intrinsic viscosity and the Huggins coefficient depend on temperature: $[\eta]$ decreases and k_1 increases with increasing temperature. The numerical values of $[\eta]$ and k_1 for all investigated IgG are presented in Tables 2 and 3. The intrinsic viscosity of investigated IgG is differentiated and only at the high temperature limit ($t = 55^\circ\text{C}$) does $[\eta]$ have similar values for human and bovine IgG. The Huggins coefficient, in turn, has similar values over the whole range of temperatures only for bovine and pig IgG. The numerical values of k_1 which are published in the literature [11,15,17,18] are very differentiated for different liquids and solutions. It follows from the fact that the intermolecular hydrodynamic interactions can be different in each case. Thus, the experimental values of k_1 for different liquids are still desirable.

4. Conclusions

The viscosity of mammalian IgG solutions at temperatures up to 55°C , over a wide range of concentrations and at pH values near the isoelectric point, may be quantitatively described by the generalized Arrhenius formula (Eq. (5)). The asymptotic form of the formula for small concentrations enables calculation of the in-

Table 3

The Huggins coefficient k_1 for all investigated IgG calculated from Eq. (8)

	<i>t</i> (°C)										
	5	10	15	20	25	30	35	40	45	50	55
Human	0.7148	0.7225	0.7302	0.7377	0.7451	0.7522	0.7592	0.7658	0.7721	0.7781	0.7836
Bovine	0.7907	0.8053	0.8203	0.8357	0.8514	0.8674	0.8837	0.9001	0.9166	0.9330	0.9494
Pig	0.8044	0.8186	0.8324	0.8456	0.8581	0.8698	0.8804	0.8897	0.8977	0.9042	0.9092

intrinsic viscosity and the Huggins coefficient. Both quantities depend on temperature: $[\eta]$ decreases and k_1 increases with increasing temperature. Analysis of the relative viscosity–concentration relation suggests that human and pig IgG molecules have almost exactly the same shape. Despite the similarities, substantial differences exist between species, especially for values of the parameters from Eqs. (2)–(4). This indicates that each mammalian IgG in solution should be separately studied in detail.

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